Faeze Mohammadi

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RESEARCH PROFILE

Computer Engineering undergraduate at the University of Tehran pursuing research in artificial intelligence, machine learning, and deep learning. I work with Dr. Ahmad Kalhor on data selection and representation evaluation, and with Dr. Khatami under Dr. Siamak Mohammadi on neural networks for side-channel analysis. My independent projects include a Separation Index representation audit, Persian poetry attribution and retrieval, and blood-cell classification. Across these projects, I develop and evaluate models through controlled experiments, baseline comparisons, and analysis of their behavior and limitations.

EDUCATION

University of Tehran - BSc student in Computer Engineering

GPA: 19.68/20. Academic rank: **1st** in the Computer Engineering entering cohort, based on the first two years of study.

Selected grades (20/20): Probability and Statistics; Discrete Mathematics; Artificial Intelligence; Data Structures; Advanced Programming; Formal Languages and Automata; Engineering Mathematics; Calculus II; Differential Equations.

National Olympiad Round 2 qualifier: Mathematics, Computer Science, and Economics & Management.

Secondary education: Six years in schools of the National Organization for Development of Exceptional Talents (NODET; SAMPAD), 2018-2024. **Diploma GPA: 19.86/20.**

CURRENT RESEARCH

Data and Representation Evaluation with SI and SmI

University of Tehran | Faculty-guided research with Dr. Ahmad Kalhor | In progress

- Working with Dr. Ahmad Kalhor on an ongoing research project involving Separation Index (SI) and Smoothness Index (SmI) for evaluating data and neural representations.
- The research covers data difficulty, quantity, diversity, and coverage; changes in network representations during training; and feature or layer removal while maintaining accuracy. Current SI benchmarking uses MNIST, Fashion-MNIST, CIFAR-10, and CIFAR-100 with controlled comparisons and separate training and validation sets.
- These studies form the foundation of a broader research project that continues to develop under Dr. Kalhor's guidance.

Neural Networks for Profiling Side-Channel Analysis

University of Tehran | Guided by Dr. Khatami under Dr. Siamak Mohammadi | In progress

- Evaluating MLP, CNN, LSTM, and a CNN with linear-attention adapter on ASCAD, using 50,000 profiling traces and 10,000 attack traces, targeting recovery of a single AES key byte.
- Analyzed multi-seed key-rank curves, checkpoint replay, and ablations. CNN/MLP and hybrid experiments recovered the target byte across three seeds; recurrent follow-ups retained unsuccessful results.
- Assessed attention ablations and seed variability in exploratory experiments; the hybrid showed mixed recovery efficiency relative to matched CNN controls.

TEACHING EXPERIENCE

Teaching Assistant / TA Supervisor - Discrete Mathematics

University of Tehran | Instructor: Dr. Siamak Mohammadi

- Serving a third term as a teaching assistant, with TA supervisor responsibilities this term. Supporting students in mathematical logic, proof techniques and induction, recurrence relations, counting, and graph theory.

Teaching Assistant - Formal Languages and Automata

University of Tehran | Instructor: Dr. Hossein Hojjat

- Supported students in understanding formal languages, grammars, and finite automata, with an emphasis on the connections between language descriptions and computational models. Helped students reason through automata constructions, formal proofs, and course problems.

TECHNICAL SKILLS

Programming: Python, C++, C, R, Verilog.
Machine Learning and Data Analysis: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, NumPy, pandas.

SELECTED PROJECTS

Persian Poetry Attribution and Search

Repository | Independent research prototype | Python, scikit-learn, Tesseract, Vosk

- Built separate author-attribution and known-item retrieval evaluations on 500 poems by five authors. Attribution used 300/100/100 train/validation/test poems, with duplicate controls and author-name masking.
- Compared word/character TF-IDF with LinearSVC; the character model achieved clean test macro-F1 of 0.9502 on 100 held-out poems. Assessed synthetic corruption using paired bootstrap intervals.
- Integrated OCR and speech recognition into a local demo; evaluated a bounded pilot of 20 rendered images, their transformed versions, and three recitation clips separately from text classification.

Separation Index Representation Audit

Repository | Independent project | Python, PyTorch, NumPy

- Audited a published SI definition on Optdigits MLPs and Fashion-MNIST CNNs, using paired seeds, fixed baselines, nearest-neighbor self-exclusion, and deterministic distance ties.
- Compared representation diagnostics with held-out classification. Reported constant-SI and negative results, baseline performance, and convergence warnings.

Blood Cell Classification with PyTorch

Repository | Python, PyTorch, torchvision, NumPy

Compared MLP and CNN models for five-class blood-cell classification using approximately matched 3.18-million-parameter budgets. Analyzed learning curves, confusion matrices, prediction confidence, and first-layer feature maps; tested responses to rotation, translation, and image occlusion.

MountainCar Reinforcement Learning with Q-Learning

Repository | Python, NumPy, Gym

Implemented tabular Q-learning from scratch for MountainCar-v0 with discretized position and velocity. Compared four exploration schedules, state-grid resolutions, and reward shaping. Evaluated policies over 100 greedy episodes and analyzed reward hacking using goal-reaching success independently of training reward.

Genetic Music Reconstruction and Minimax Checkers

Repository | Python, NumPy, librosa, Pygame

Reconstructed melodies with a genetic algorithm on FFT-derived frequency sequences, using selection, crossover, mutation, and elitism. Built Minimax and Alpha-Beta Checkers agents; compared search-depth runtimes and the effects of population size, generations, and mutation rate on reconstruction fitness.

Supervised Learning from Scratch

Repository | Python, NumPy, pandas, scikit-learn

Implemented binary decision trees and multiclass softmax regression from scratch. Compared Naive Bayes, Random Forest, Logistic Regression, and KNN baselines; analyzed feature engineering, class imbalance, and scaling. Fitted preprocessing on training data and evaluated precision, recall, F1, and confusion matrices.

Additional Projects in Probabilistic Inference and Search

Bayesian robot localization (R) with a discrete Bayes filter under sensor and motion uncertainty; Naive Bayes spam classification (R); and pathfinding with BFS, DFS, IDS, A*, and Weighted A*.

SELECTED COURSES AND CERTIFICATES

Machine Learning Specialization - DeepLearning.AI / Stanford Online via Coursera
CS50's Introduction to Artificial Intelligence with Python - Harvard CS50
Cryptography I - Stanford Online via Coursera
Vision Language Models Bootcamp - OpenCV University
PyTorch Bootcamp - OpenCV University
OpenCV Bootcamp - OpenCV University
TensorFlow-Keras Bootcamp - OpenCV University
Writing in the Sciences - Stanford Online via Coursera
Additional certificates: View all credentials

ADDITIONAL ACADEMIC WRITING

A Study of the Higgs Boson Particle. Sole-authored literature review manuscript; available upon request. Synthesized scientific literature into a structured technical review.